

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical : Using C language in Ubuntu

Practical Sheet No: Lab_1

Student ID	IT24102099
Student Name	Mathuppriya Naguleswaran
Campus/ Center name	SLIIT – Northern UNI
Specialization	Software Engineering
Batch No	1

Ubuntu Environment and Terminal

❖ **Terminal = It is a place where you type commands. (C programs are usually compiled and run through the terminal.)**

mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox : ~ \$

- mathuppriya-nakuleswaran = Current name
- @mathuppriya-nakuleswaran-VirtualBox = Computer name
- ~ = Current folder, usually home directory
- \$ = you can type a command

❖ **Follow of the work.**

Create a C file => Write code => Save code => Compile code => Execute code

1. Create a C file : [**nano** first.c] here, nano is a text editor.
2. Write code : It includes standard input-output library.
3. Save code : Ctrl + O then, exit with Ctrl + X
4. Compile code : gcc -o first first.
 - gcc = GNU C compiler
 - -o first = create output file named first
 - First.c = source code file
5. Execute code : ./first

❖ **Instructions**

- printf () : used to display output on the screen.
- main () : C program start execution from the main () function.
- #include <stdio.h> : we use this because printf () is defined in that library.
- scanf () : keyboard input using scanf.
- 50/100.0 gives 0.5, not 0 because 100.0 is a decimal number, C treats the division as floating-point division
- %.2f : Display the result with 2 decimal places.

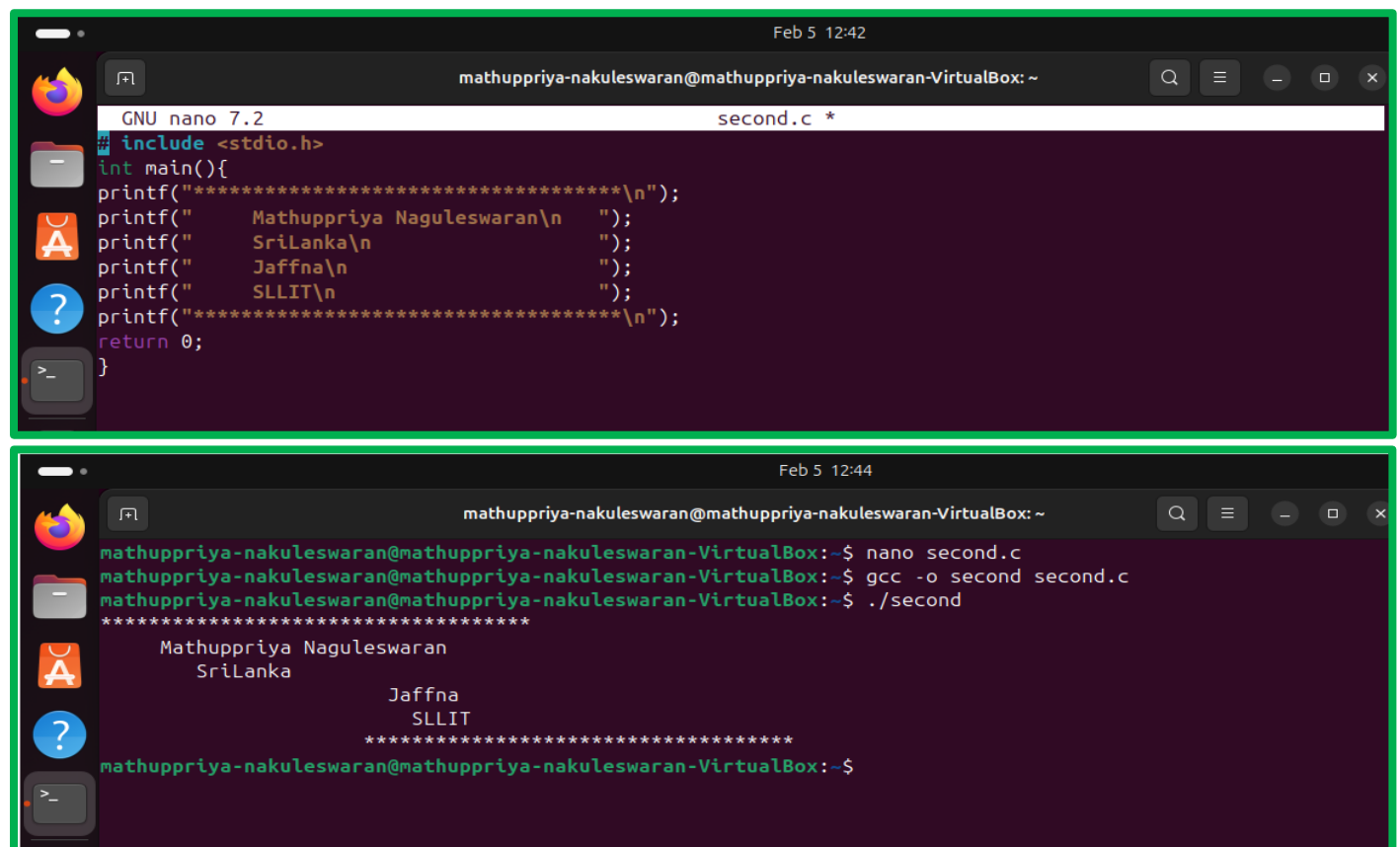
About %d

%d is a format specifier used in C to print or read an integer value. It tells the program that the value is of type integer.

About scanf()

scanf() is used to take input from the keyboard. It reads the value entered by the user and stores it in a variable using its memory address.

❖ Activity 1 – Display Name, Country, University use use with printf ()



The figure consists of two screenshots of a terminal window. The top screenshot shows the GNU nano 7.2 editor editing a file named 'second.c'. The code inside the file is as follows:

```
GNU nano 7.2 second.c *
#include <stdio.h>
int main(){
printf("*****\n");
printf("    Mathuppriya Naguleswaran\n");
printf("    SriLanka\n");
printf("    Jaffna\n");
printf("    SLLIT\n");
printf("*****\n");
return 0;
}
```

The bottom screenshot shows the terminal after compilation and execution. The commands entered are:

```
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox: ~
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ nano second.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ gcc -o second second.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ ./second
```

The output of the program is displayed as follows:

```
*****
    Mathuppriya Naguleswaran
    SriLanka
    Jaffna
    SLLIT
*****
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$
```

Figure 1.1 : Create C file then compile ,execute it.

❖ Activity 2 – Data types and variables in C

Variable = It is like a container that stores data in memory.

Int a = 10;

- This means,
- Create a memory space
- Name it a
- Store value 10 inside it

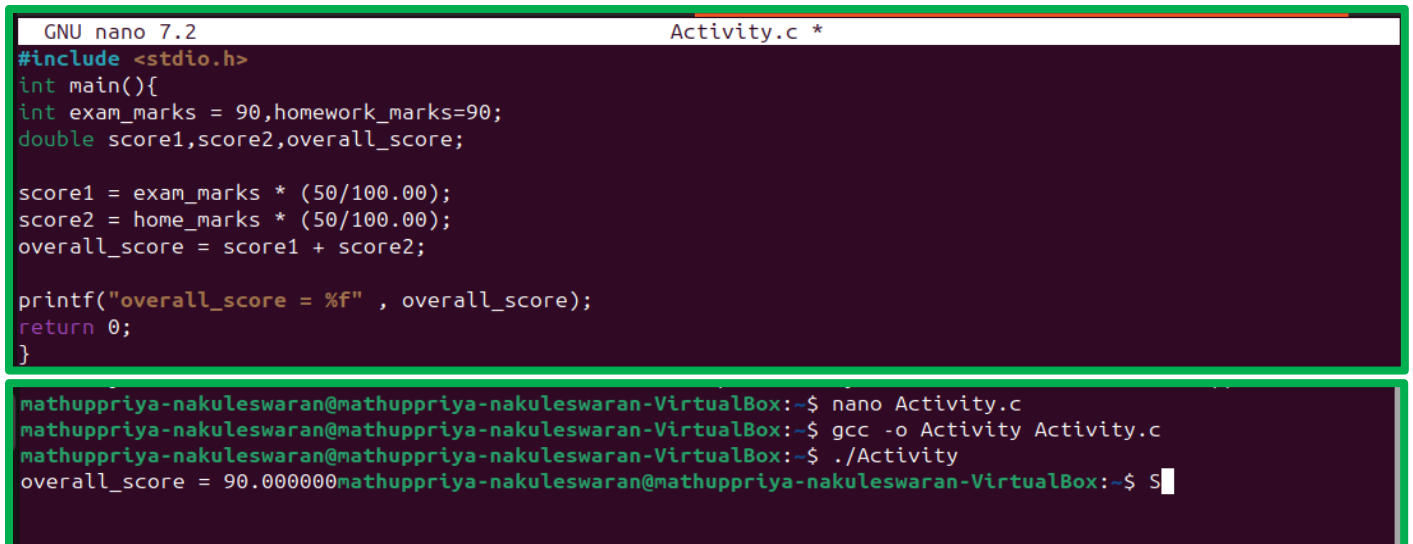
Variable names are case sensitive.

int : Used to store whole number [EX: `int age = 20;`]

float : Used to store decimal numbers [Ex: `float marks = 75.5;`]

double : Also Used to store decimal numbers but, more precise than float.

char : Used to store a single character [EX: `char grade = 'A';`]



```
GNU nano 7.2 Activity.c *
#include <stdio.h>
int main(){
int exam_marks = 90, homework_marks=90;
double score1, score2, overall_score;

score1 = exam_marks * (50/100.00);
score2 = homework_marks * (50/100.00);
overall_score = score1 + score2;

printf("overall_score = %f" , overall_score);
return 0;
}

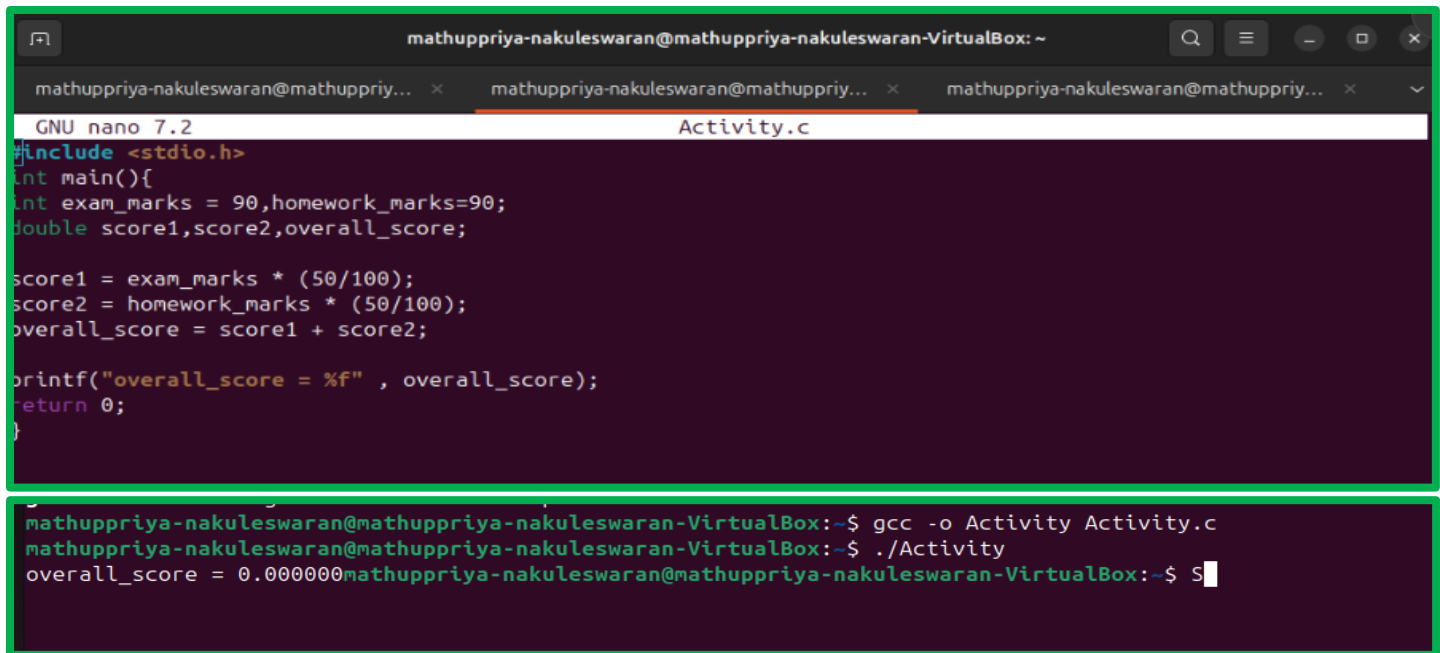
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ nano Activity.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ gcc -o Activity Activity.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ ./Activity
overall_score = 90.000000mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ S
```

Figure 1.1 : Create C file then compile ,execute it.

Explanation of the code : Here, im using fixed numbers to assign and define the variables. The variables are exam_marks it assign with 90 and likewise homework_marks = 90. To calculate above values im used some other variables to store those values, those are score1, score2, overall_score. For get the overall marks we want to calculate both for 50% then add up those.

- %f : The **format specifier** for float input
- &exam_marks : The memory address where the value should be stored.
- Use float instead of int because the calculation may produce decimal values.
- Why we use &exam_marks : because scanf() needs the memory address to store the input. (don't forget & in scanf().)

❖ Activity 3 – Difference between integer division & floating division



```
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox: ~
GNU nano 7.2 Activity.c
#include <stdio.h>
int main(){
int exam_marks = 90, homework_marks=90;
double score1, score2, overall_score;

score1 = exam_marks * (50/100);
score2 = homework_marks * (50/100);
overall_score = score1 + score2;

printf("overall_score = %f" , overall_score);
return 0;
}

mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ gcc -o Activity Activity.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ ./Activity
overall_score = 0.000000mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$
```

Figure 1.2 : Create C file then compile ,execute it.

50 / 100 uses integer division,

In C language :

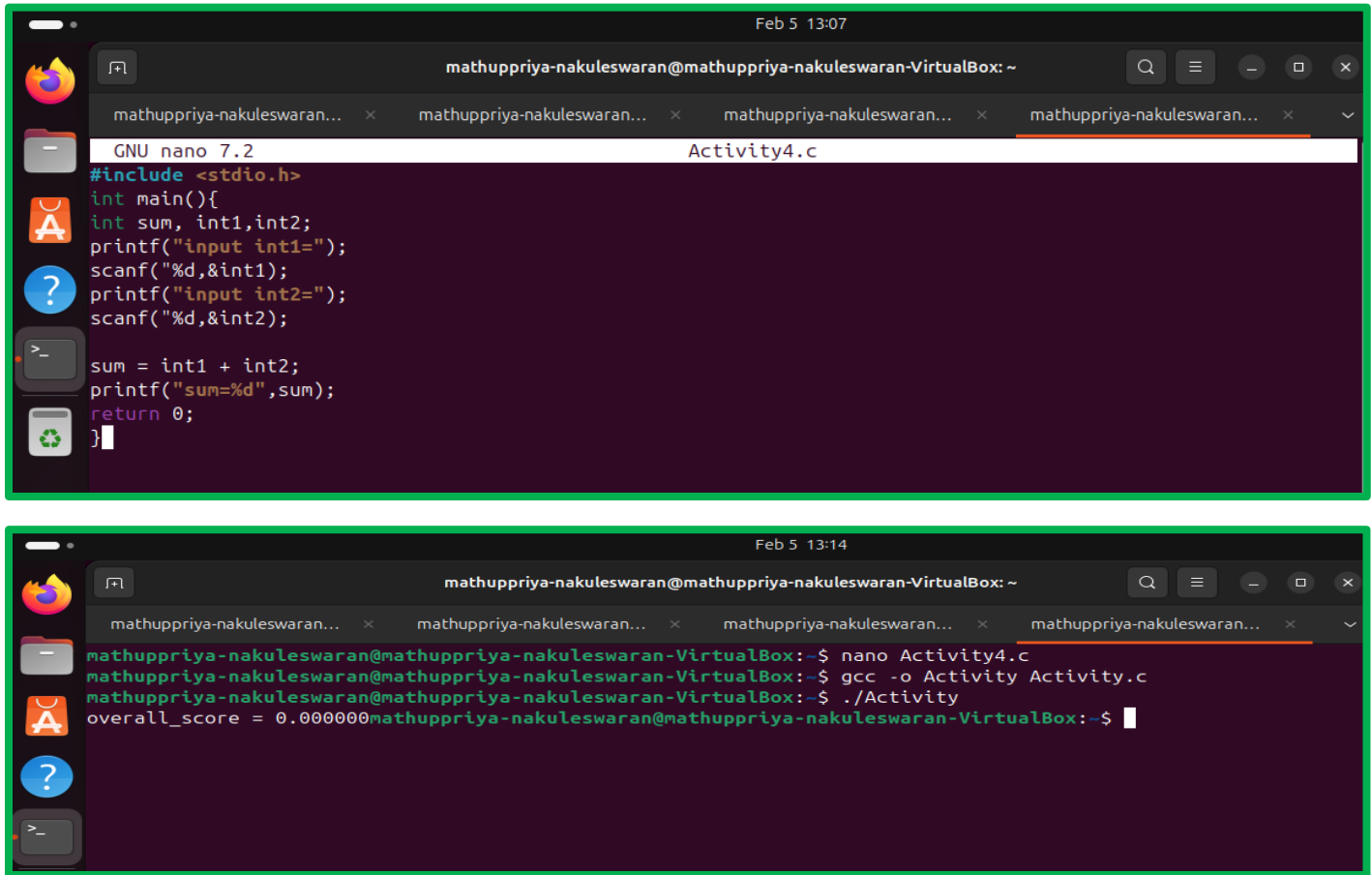
- 50 is an integer
- 100 is an integer
- Integer division does not keep decimal part.
- So, $50 / 100 = 0$
- Then , $\text{score1} = \text{exam_marks} * 0 = 0.00$

Explanation of the code : Here, the output becomes zero because the expression $50 / 100$ performs integer division in C. Since both 50 and 100 are integers, the decimal part is removed and the result becomes 0 instead of 0.5 Therefore, when the exam and homework marks are multiplied by this value, both calculated scores become zero, and the final overall score is also zero.

Integer division : Division between integers where the decimal part is removed.

- $50 / 100 = 0$
- $50 / 100.0 = 0.5$

❖ Activity 4 – Read 2 integer values from the user & calculate their sum.



```
GNU nano 7.2 Activity4.c
#include <stdio.h>
int main(){
int sum, int1,int2;
printf("input int1=");
scanf("%d",&int1);
printf("input int2=");
scanf("%d",&int2);

sum = int1 + int2;
printf("sum=%d",sum);
return 0;
}
```

```
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox: ~
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ nano Activity4.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ gcc -o Activity Activity.c
mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$ ./Activity
overall_score = 0.000000mathuppriya-nakuleswaran@mathuppriya-nakuleswaran-VirtualBox:~$
```

Figure 1.3 : Create C file then compile ,execute it.

Explanation of the code : Create 3 integer variables ,which are int1, int2, sum.Then ask the user to get the values for those variables and store it. Then add both user inputs and display the summation of those inputs.

Scanf () : reads input from the keyboard.

EX :Int marks;

Scanf(“%d”, &marks);

- Wait for user input
- Expect an integer because %d is used
- Store it in marks

%d used for integer values as input/output.

Why &marks : scanf() needs the address of the variable, not just the variable name.(without & the program may give an error or behave incorrectly.